

Sets

- **Exercise 1**

Suppose $A, B, D \subseteq X$. Prove the following properties

1. $(A \setminus B)^c = A^c \cup B$.

Proof

$$\begin{aligned} x \in (A \setminus B)^c &\iff x \notin (A \setminus B) \iff \text{It is false that } x \in A \setminus B \\ &\iff \text{It is false that } x \in A \wedge x \notin B \\ &\iff \text{It is true that } \neg(x \in A \wedge x \notin B) \\ &\iff x \notin A \vee x \in B \iff x \notin A \text{ or } x \in B \\ &\iff x \in A^c \cup B \end{aligned}$$

Thus $(A \setminus B)^c = A^c \cup B$.

2. $A \cap (B \setminus D) = (A \cap B) \setminus (A \cap D)$.

Proof

$$\begin{aligned} x \in A \cap (B \setminus D) &\iff x \in A \wedge x \in B \setminus D \\ &\iff x \in A \wedge x \in B \wedge x \notin D \\ &\iff (x \in A \wedge x \in B) \wedge (x \in A \wedge x \notin D) \\ &\iff x \in A \cap B \wedge x \notin A \cap D \\ &\iff x \in (A \cap B) \setminus (A \cap D) \end{aligned}$$

Thus $A \cap (B \setminus D) = (A \cap B) \setminus (A \cap D)$.

- **Exercise 2**

Let A and B be two sets. Prove that $\mathcal{P}(A) \cup \mathcal{P}(B) \subseteq \mathcal{P}(A \cup B)$.

- **Proof 2.1**

Suppose

$$\begin{aligned} X \in \mathcal{P}(A) \cup \mathcal{P}(B) &\implies X \in \mathcal{P}(A) \text{ or } X \in \mathcal{P}(B) \\ &\iff X \subseteq A \text{ or } X \subseteq B \\ &\implies X \subseteq A \subseteq A \cup B \text{ or } X \subseteq B \subseteq A \cup B \\ &\implies \text{In any case, } X \subseteq A \cup B \\ &\implies X \in \mathcal{P}(A \cup B) \end{aligned}$$

Thus $\mathcal{P}(A) \cup \mathcal{P}(B) \subseteq \mathcal{P}(A \cup B)$.

- **Exercise 3**

Suppose $A_n = \left[-\frac{1}{n}, 1 + \frac{1}{n}\right]$ for $n \in \mathbb{N}$, find $\bigcup_{n \in \mathbb{N}} A_n$, $\bigcap_{n \in \mathbb{N}} A_n$, $\bigcup_{n \in \mathbb{N}} A_n^c$, and $\bigcap_{n \in \mathbb{N}} A_n^c$.

- **Solution 3.1**

As n increases, the intervals A_n become nested, meaning $A_1 \supseteq A_2 \supseteq A_3 \supseteq \dots$

For $n = 1$, we have $A_1 = [-1, 2]$.

The **union** is equal to the largest interval

$$\bigcup_{n \in \mathbb{N}} A_n = [-1, 2]$$

The **intersection** approaches the limit points 0 and 1 from the outside, including them

$$\bigcap_{n \in \mathbb{N}} A_n = [0, 1]$$

By **De Morgan's laws**, the union of the complements is the complement of the intersection

$$\bigcup_{n \in \mathbb{N}} A_n^c = \left(\bigcap_{n \in \mathbb{N}} A_n \right)^c = (-\infty, 0) \cup (1, \infty)$$

Similarly, the intersection of the complements is the complement of the union

$$\bigcap_{n \in \mathbb{N}} A_n^c = \left(\bigcup_{n \in \mathbb{N}} A_n \right)^c = (-\infty, -1) \cup (2, \infty)$$